Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Theory of Computation (UCS701/ULC703) – End Semester Examination
Date: 13/05/2026 & Time: 2:00PM MM: 50 & MT: 180 Min

List of Faculties: Dr. Chinmaya Panigrahy (C2), Dr. Amrita Kaur (C1), Dr. Aditi Sharma (C6, P1), Dr. Vibha Jain (C3, Q2), Dr. Nidhi Chakravarty (C7, Q1), and Dr. Nitin Arora (C4, C5)

Attempt any five of the six questions only. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No		Questions	Marks	CO	BL
Q1.	(a)	While simplifying a given CFG, under the removal of useless symbols section, Devansh first removed the non-reachable symbols and then eliminated the non-generating symbols. Will he always get a simplified CFG? Justify your answer with a suitable example.	2	CO2	L2
	(b)	Consider the following CFG. $E \rightarrow I \mid E+E \mid E*E \mid (E)$ $I \rightarrow a \mid b \mid c$ (i) Prove that the above CFG is ambiguous. (ii) Remove ambiguity from the above CFG. (iii) Remove left-recursion from the given CFG.	5	CO2	L4
	(c)	Design a CFG to generate all the strings over {a,b} having unequal number of a's and b's. Clearly mention each component of the CFG.	3	CO2	L3
Q2.	(a)	Consider the language, $L = \{0^n 1^{2n} \mid n>0\}$. For the L, (i) Write a logic to design a PDA for L. (ii) Draw a flow diagram for the respective PDA by applying the logic of (i). (iii) Design a PDA to accept L through final state. (iv) Write the tuple-representation of the PDA constructed in (iii). (v) Show the sequence of ID's the PDA will move through while processing the string 001111. (vi) Convert the PDA constructed in (iii) to accept L through empty stack.	8	CO2	L3
	(b)	Prove that the context-free languages are not closed under complement operation.	2	CO2	L2
Q3.	(a)	Convert the following context-free grammar into Greibach normal form. $S \rightarrow AA \mid a$ $A \rightarrow SS \mid b$	4	CO2	L4
	(b)	Use the CYK algorithm to determine whether the string abaab is in the language generated by the following grammar. $S \rightarrow AB \mid SS \mid a$ $A \rightarrow BS \mid CD \mid b$ $B \rightarrow DD \mid b$ $C \rightarrow DE \mid a \mid b$ $D \rightarrow a$ $E \rightarrow SS$	4	CO2	L3
	(c)	Show that the language, $L = \{0^X \mid X=n^2 \mid n > 0\}$ is not context-free using the Pumping lemma for context-free languages.	2	CO2	L3

Q4.	**(a)**	Consider the following languages. L1={x | x does not contain substring 101, x ∈ (0,1)*} L2={x | x ends with 01, x ∈ (0,1)*} Use the finite automata of L1 and L2 to derive the finite automata recognizing the following languages. (i) L1 ∪ L2 (ii) L1 ∩ L2 (iii) L1 − L2	**5**	**CO1**	**L4**
	(b)	Design a left-linear and right-linear grammar for the following regular expressions. (i) 10(100+011+010)* (ii) (100+011+010)*10 (iii) 10(100+011+010)*01	**3**	**CO1**	**L3**
	(c)	List all the strings of length 4 or less in the regular set represented by the regular expression, (0*1+1*0)0*.	**2**	**CO1**	**L3**
Q5.	**(a)**	Design a Turing machine for the language, L = $\{0^X \mid X=2^n$, where $n \geq 0\}$, by writing the logic. Show how the above Turing machine will process the string, w = 000 by mentioning the sequence of IDs. Represent the above Turing machine through a suitable transition table and tuple-representation.	**7**	**CO3**	**L3**
	(b)	Design a two-stack PDA for the language, L = $\{a^n b^n c^n \mid n \geq 0\}$. Give a tuple-representation of the above two-stack PDA.	**3**	**CO3**	**L3**
Q6.	**(a)**	Design a Turing machine to right shift the binary input by one position and # will be placed at the first position. What will be the first and last IDs of the above Turing machine? Show how the content of the input tape will be updated while shifting the string 101.	**4**	**CO3**	**L3**
	(b)	Write down the logic for designing a Turing machine for the division of two positive integers. Show the processing of a sample input.	**3**	**CO3**	**L4**
	(c)	Simplify the following CFG. S → ASB | ε A → aAS | a B → SbS | A | bb	**3**	**CO2**	**L3**

.....All the Best.....